

## ENGINE LEVEL — IS THE GEOMETRY RIGHT?

### 1 Line-of-sight engine

profile sampling · curvature + refraction

5 switchable defects, injected one at a time



### 2 Benchmark

15 terrain cases — expectations derived from the critical distance, not intuition

4 behavioural properties, random terrain



### 3 Mutation analysis

the whole benchmark, run against each deliberately broken engine



### 4 4 defects killed · 1 equivalent

equivalence proven by a paired sweep, not assumed



### Diagnostics for any real study

D1 null placements touching water = 0



ONLY A VALIDATED ENGINE AND A DIAGNOSED DESIGN TOUCH REAL TERRAIN

## DESIGN LEVEL — IS THE INFERENCE RIGHT?

### 1 Synthetic landscape

anisotropic ridges · a lake flooded to a constant, exactly as real DEMs build lakes



### 2 Ground truth by construction

S0 exchangeable → p exactly uniform

S1 best of K → known theoretical power (ceiling 85 %)



### 3 Paired contrasts

correct · water unmasked · tight crop  
same data — only the defect differs



### 4 Calibration · power · mechanism

each defect measured against an exact expectation, not an assumption



D2 orientation coverage = 100 %